Journal of Power Sources 248 (2014) 77–82

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Short communication

Synthesis of novel organic-ligand-doped sodium bis(oxalate)-borate complexes with tailored thermal stability and enhanced ion conductivity for sodium ion batteries

Chunhua Ge, Lixia Wang, Lili Xue, Zhong-Shuai Wu, Hehe Li, Zailin Gong, Xiang-Dong Zhang*

College of Chemistry, Liaoning University, Shenyang 110036, PR China

HIGHLIGHTS

- A series of new-type sodium boron salts are prepared by solid-state reaction.
- The organic-ligand was doped to change solubility and thermal stability of the boron slats.
- The resulting sodium boron salts may be used as electrolytes in sodium ion batteries.

ARTICLE INFO

Article history:
Received 4 July 2013
Received in revised form
9 September 2013
Accepted 11 September 2013
Available online 25 September 2013

Keywords:
Sodium boron salts
Sodium ion battery
Electrolyte
Electrochemistry
Thermal stability
Ion conductivity

ABSTRACT

A series of novel organic-ligand-doped sodium bis(oxalate)-borate complexes, including sodium bis[salicylato(2-)]-borate (NBSB), sodium[salicylato benzenediol]borate (NBDSB), sodium bis[oxalate]-borate (NBOB) and its derivatives NBOB$(C_2H_2O_4)_{0.2}$, NBOB$(C_7H_4O_3)_{0.2}$, NBOB$(C_7H_4O_3)_{0.6}$, NBOB$(C_6H_6O_2)_{0.15}$ and NBOB$(C_6H_6O_2)_{0.3}$ fabricated by solid-state reaction are firstly developed as new-type electrolytes for sodium ion batteries. These resulting sodium boron salts possess good solubility in an abroad range of organic solvents (such as PC, AN, DMF, PC + AN, PC + DMF), tailored thermal stability from 300 to 353 °C, improved ion conductivity ($>1 \times 10^{-3}$ S cm^{-1}), environmental friendliness and low cost. Therefore, we believe that these new-type sodium boron salts show great potential as a new class of electrolyte for high-performance sodium ion batteries.

1. Introduction

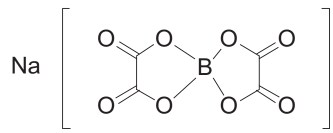
With increasing concerns on global warming, decrease of the fossil fuels, high-cost lithium and sustainable resources [1–6], sodium-ion batteries that use sodium-ions as a way to store power in a compact system have been emerged as one of the most promising candidates of reusable batteries for portable electronics, electrical vehicles, and other stationary energy storage systems. Although this type of battery is still in a fundamental research stage but is anticipated to be a cheaper, more durable technique to store energy than commonly used lithium-ion batteries, due to the rich resource of sodium metal in the earth (6th most abundant element), and the similar chemical properties of lithium as well as easy operation of sodium ion batteries at room temperature [7–12].

So far, several interesting sodium salts have been applied for sodium ion batteries, including $NaPF_6$, $NaSbF_6$, $NaAsF_6$, $NaClO_4$, $NaBF_4$, $NaSO_3CF_3$, $NaCF_3CO_2$ and $Na(CH_3)C_6H_4SO_3$ [13]. However, $NaAsF_6$ and $NaSbF_6$ possess toxicity, and pollute the environment. $NaClO_4$ is readily explosive and unsafe. $NaBF_4$ and $NaPF_6$ are incompatible with cathode materials and sensitive to moisture due to the low thermal stability. As to $NaSO_3CF_3$, $NaCF_3CO_2$ and $Na(CH_3)C_6H_4SO_3$ electrolytes, their conductivity values are relatively low. Therefore, it is necessary to develop new, environmentally friendly, safe, high conductive, and low-cost sodium salts as electrolytes for sodium ion batteries [14–23].

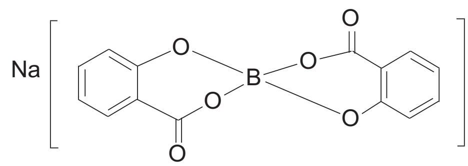
It was reported that lithium bis(oxalate) borate (LiBOB) is a most promising potential salt for lithium ion battery electrolytes [24–26]. This is because LiBOB possesses excellent thermal stability at

* Corresponding author. Tel.: +86 24 62202053; fax: +86 24 62202380.
E-mail address: xd623@sina.com (X.-D. Zhang).

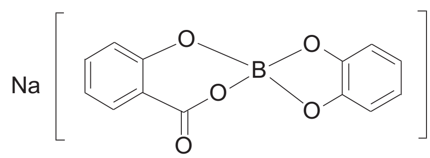
0378-7753/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jpowsour.2013.09.044



NBOB



NBSB



NBDSB

Scheme 1. Structures of NBOB, NBSB and NBDSB.

high temperature (such as 70 °C) that no other lithium salt has (For example, $LiPF_6$ is the most widely used electrolyte salt but totally decomposed at 80 °C) [24–26]. This high-temperature stability of LiBOB electrolyte is crucial for the use and operation safety of lithium ion batteries for electrical vehicles when the ambient temperature in summer rises up to 70 °C [27,28]. From a structure point of view, the structure of sodium bis[oxalate]-borate is very similar to that of LiBOB, therefore, it may be reasonably predicted that the sodium bis[oxalate]-borate (NBOB) salt may have the similar high-temperature stability of LiBOB and great potential as electrolyte for sodium ion battery [27].

In this work, we first developed a new facial strategy to fabricate a series of novel organic-ligand-doped sodium bis(oxalate)-borate complexes, including NBOB and $NBOB(C_6H_6O_2)_{0.3}$ sodium bis[salicylato(2)]-borate (NBSB), sodium[salicylato benzenediol]borate (NBDSB, Scheme 1), and five NBOB derivatives of $NBOB(C_2H_2O_4)_{0.2}$, $NBOB(C_7H_4O_3)_{0.2}$, $NBOB(C_7H_4O_3)_{0.6}$, $NBOB(C_6H_6O_2)_{0.15}$ by solid-state reaction, and demonstrated these resulting sodium boron salts possess good solubility in different organic solvents, enhanced thermal stability and improved ion conductivity for sodium ion batteries, which are in good agreement with the requirements of commercially available electrolytes for battery [19,29–33]. Notably, these new-type sodium boron salts show great potential as a new type of electrolyte for sodium ion battery because of their environmental friendliness, cheap and high thermal stability and ion conductivity. Moreover, we believe that, through further optimization of the type of organic-ligand dopants and suitable solvents, the solubility and conductivity of electrolytes from these sodium bis(oxalate)-borate complexes can be further improved, which should be useful for the applications in sodium ion batteries and other energy storage devices.

2. Experimental

NBOB, NBSB, NBDSB and the five derivates of NBOB doped with organic acids and pyrocatechol ($NBOB(C_2H_2O_4)_{0.2}$, $NBOB(C_7H_6O_3)_{0.2}$, $NBOB(C_7H_6O_3)_{0.6}$, $NBOB(C_6H_6O_2)_{0.15}$ and $NBOB(C_6H_6O_2)_{0.3}$) were synthesized by solid-state reaction, and the reaction feed ratio is the ratio of reaction materials.

$$2C_2H_2O_4\cdot 2H_2O + NaOH + H_3BO_3 \rightarrow NaBO_4(C_2O_2)_2 \tag{1}$$

$$2C_7H_6O_3 + NaOH + H_3BO_3 \rightarrow NaBO_4(C_7H_4O)_2 \tag{2}$$

$$C_7H_6O_3 + NaOH + H_3BO_3 + C_6H_6O_2 \rightarrow NaBO_4(C_7H_4O)(C_6H_4) \tag{3}$$

$$2C_2H_2O_4\cdot 2H_2O + NaOH + H_3BO_3 + nC_2H_2O_4\cdot 2H_2O \rightarrow NaBO_4(C_2O_2)_2(C_2H_2O_4)_n \tag{4}$$

$$2C_2H_2O_4\cdot 2H_2O + NaOH + H_3BO_3 + nC_7H_6O_3 \rightarrow NaBO_4(C_2O_2)_2(C_7H_6O_3)_n \tag{5}$$

$$2C_2H_2O_4\cdot 2H_2O + NaOH + H_3BO_3 + nC_6H_6O_2 \rightarrow NaBO_4(C_2O_2)_2(C_6H_6O_2)_n \tag{6}$$

NBOB was synthesized by solid-state reaction according to reaction (1), oxalic acid, boric acid and sodium hydroxide were mixed in a 2:1:1 molar ratio and refined by dry grinding. The synthesis process included two steps, the reaction temperature and reaction time were ~110 °C for 6 h and then 240 °C for 6 h. The resulting powder was washed with ethanol several times, and dried in a vacuum oven at 60 °C for 4 h. After that, the sample was examined by element analysis for C and H contents, which were measured as C 22.69% and H 0.047%. These data were close to the calculated values of C 22.90% and H 0.00% for NBOB.

NBSB was synthesized by solid-state reaction according to reaction (2), salicylic acid, boric acid and sodium hydroxide were mixed in a 2:1:1 molar ratio, synthesized and purified in a similar procedure for NBOB. The element analysis of NBSB was C 54.79% and H 2.474%.

NBDSB was synthesized by solid-state reaction according to reaction (3), salicylic acid, boric acid, sodium hydroxide and pyrocatechol were mixed in a 1:1:1:1 molar ratio, synthesized and purified in a similar procedure for NBOB. The element analysis of NBDSB showed C 53.83% and H 2.523%.

$NBOB(C_2H_2O_4)_{0.2}$ was synthesized by solid-state reaction according to reaction (4), the element analysis showed C 22.92% and H 0.178%.

$NBOB(C_7H_6O_3)_{0.2}$ and $NBOB(C_7H_6O_3)_{0.6}$ were synthesized by solid-state reaction according to reaction (5), the element analysis of the $NBOB(C_7H_6O_3)_{0.2}$ had C 27.12% and H 0.593%, the element analysis of $NBOB(C_7H_6O_3)_{0.6}$ showed C 34.36% and H 0.852%.

$NBOB(C_6H_6O_2)_{0.15}$ and $NBOB(C_6H_6O_2)_{0.3}$ were synthesized by solid-state reaction according to reaction (6), the element analysis of the $NBOB(C_6H_6O_2)_{0.15}$ had C 26.09% and H 0.428% while $NBOB(C_6H_6O_2)_{0.3}$ showed C 28.67% and H 0.649%.

Fourier transformed infrared (FTIR) analysis of eight sodium borate salts incorporated into KBr disks was performed using an FTIR spectrometer (Nicolet AVATAR 360, USA) over the range 400–4000 cm^{-1}. X-ray diffraction (XRD) patterns were obtained by a D-rmax diffractometer (Cu Ka, 30 kV). The morphology of NBOB, NBSB and NBDSB were observed by scanning electron microscopy (FEI QUANTA200FEG). Thermogravimetric (TG) analysis of eight sodium borate salts were carried out using a thermogravimetric analyzer (METTLER TGA/SDTA851e), using a sample of about 2–5 mg, with a heating rate of 10 °C min^{-1} over the temperature range of 25–700 °C under the dry flow of N_2 at the rate of 20 ml min^{-1}. The preparation of the electrolyte solutions and the cell assembly were carried out in a glove box (EX29-007) filled with argon. We used the conductivity meter (DDS-11A Shanghai Leici) to measure ionic conductivities. Conductivity meter was directly connected with platinum black electrodes, then the platinum black electrode inserted into the solution, and reading the ionic conductivity values.

3. Results and discussion

The organic-ligand-doped sodium bis(oxalate)-borate complexes of NBOB, NBSB, NBDSB and the five NBOB derivates of NBOB$(C_2H_2O_4)_{0.2}$, NBOB$(C_7H_6O_3)_{0.2}$, NBOB$(C_7H_6O_3)_{0.6}$, NBOB$(C_6H_6O_2)_{0.15}$ and NBOB$(C_6H_6O_2)_{0.3}$ synthesized by solid-state reaction were carefully characterized by FT-IR spectrum, XRD, SEM, TGA and four-point probe method.

Fig. 1a shows the FT-IR spectrum of NBOB, NBSB and NBDSB. For NBOB, typical characteristic peaks appear at 1820 cm^{-1} and 1776 cm^{-1} (C=O vibration coupling), 1327 cm^{-1} (C–O–B–O–C stretching vibration), 1091 cm^{-1} and 991 cm^{-1}(B(4)–O symmetric stretching vibration and anti-symmetric stretching vibration). As to NBSB, the typical absorption peaks of C=O stretching vibration (1690 cm^{-1}), B(4)–O symmetric stretching vibration (1146 cm^{-1}) and anti-symmetric stretching vibration (756 cm^{-1}), benzene ring absorption peaks at (1477 cm^{-1}, 695 cm^{-1}) are attributed to its characteristic peaks. Similarly, the main absorption peaks of NBDSB were clearly observed at 1689 cm^{-1} (C=O stretching vibration), 1353 cm^{-1} (C–O–B–O–C stretching vibration), 1146 cm^{-1} and 755 cm^{-1} (B(4)–O symmetric stretching vibration and anti-symmetric stretching vibration), 1478 cm^{-1} and 695 cm^{-1} (benzene ring). These results demonstrated the successful synthesis of NBOB, NBSB and NBDSB.

Fig. 1b shows the FT-IR spectrum of NBOB and NBOB derivates doped with five organic acids and pyrocatechol. For NBOB$(C_2H_2O_4)_{0.2}$, the vibration frequency at 1776 and 1643 cm^{-1} is assigned to C=O and C=C vibration modes of acetate group. The intensity enhancement of the peaks in C=O and C=C band vibration is sufficient to prove the doping of oxalic acid in NBOB. Meanwhile, the enhancement of the C=O vibration coupling (1819 m^{-1} and 1779 cm^{-1}), C=O stretching vibration (1687 cm^{-1}), benzene ring in the salicylic acid (1614 m^{-1}, 1480 cm^{-1}, 754 cm^{-1} and 712 cm^{-1}) for NBOB$(C_7H_6O_3)_{0.2}$, and C=O vibration coupling (1824 m^{-1} and 1776 cm^{-1}), C=O stretching vibration (1689 cm^{-1}), and benzene ring in the salicylic acid (1615 m^{-1}, 1477 cm^{-1}, 755 m^{-1} and 693 cm^{-1}) for NBOB$(C_7H_6O_3)_{0.6}$ were also clearly observed due to the introduction of salicylic acid. Similarly, the corresponding characteristics peaks of NBOB$(C_6H_6O_2)_{0.15}$ definitely appear at 1642 m^{-1}, 1478 cm^{-1} and 776 cm^{-1} (benzene ring in the pyrocatechol) because of the pyrocatechol doping. Regarding NBOB$(C_6H_6O_2)_{0.3}$, three peaks at 1642 m^{-1}, 1477 cm^{-1} and 775 cm^{-1} observed correspond to benzene ring in the pyrocatechol.

Fig. 2a shows the XRD patterns of NBOB, NBSB and NBDSB. The XRD data of NBOB, NBSB and NBDSB were simulated using the well-known program ITO [34], and the observed peak positions are good agreement with the calculated peak positions. For NBOB, the peaks at 21.96° (200), 22.82° (002), and 37.34° (113) are the three strongest peaks, which are sharp and clear confirm the as-prepared material is crystalline structure and pure-phase NBOB [30]. The monoclinic unit cell parameters for NBOB with Orthorhombic group (*Cmcm*) were calculated, a = 8.12756 Å, b = 10.3399 Å, c = 7.82108 Å, which is identical to the reported literature [30]. Further, the peaks at 17.66, 24.66 and 25.76° for NBSB are the three strongest peaks with indexing number of (011), (211) and (20). These suggest the resulting NBSB is crystalline phase. The crystal parameters, a = 12.7330 Å, b = 7.2450 Å, c = 7.2680 Å, were calculated. As to NBDSB, the characteristics peaks display at 17.72° (11), 19.02° (002), 20.82° (12), 24.64° (112), and 25.78° (21), indicting the successful fabrication of NBDSB with an integrated crystalline pure phase. The calculated cell parameters were obtained, a = 13.254 Å, b = 8.527 Å, c = 9.848 Å.

Fig. 2b shows the XRD patterns of NBOB derivates. The XRD of NBOB$(C_2H_2O_4)_{0.2}$ exhibits seven intense peaks center at 2θ angle of 13.94, 17.12, 21.92, 26.76, 31.68, 37.34 and 41.5° due to the crystalline nature of the salt. Compared to NBOB, the location of peaks is the same, only the strength of the peaks is different which shows that the functional groups of NBOB$(C_2H_2O_4)_{0.2}$ and NBOB are the same after the doping of oxalic acid.

The XRD patterns of NBOB$(C_7H_6O_3)_{0.2}$ and NBOB$(C_7H_6O_3)_{0.6}$ exhibit eight intense peaks centered at 2θ angle of 13.92, 17.56, 21.92, 24.56, 28.72, 34.24, 37.32 and 41.46°. Compared to NBOB, the peak positions are the same but the intensity of peaks is varying, then the location of three peaks is shifted. Besides, the typical characteristic peaks such as (10), (200) and (20) of NBSB appear in the NBOB$(C_7H_6O_3)_{0.2}$ and NBOB$(C_7H_6O_3)_{0.6}$, and is more pronounced in NBOB$(C_7H_6O_3)_{0.6}$, which suggested that the resulting product is the mixture of NBOB and NBSB phases. In the case of NBOB$(C_6H_6O_2)_{0.15}$ and NBOB$(C_6H_6O_2)_{0.3}$, both of them exhibit seven intense peaks centered at 13.94, 17.12, 21.92, 26.76, 31.68, 37.34 and 41.5°. In comparison with NBOB, the locations of their peaks are the same, only the intensity of the peaks decrease.

The morphology of NBOB, NBSB and NBDSB is measured by SEM, shown in Fig. 3. It was seen that the NBOB sample is composed of irregular micro-particles with size in micrometers. NBSB consists of sheet-like structure with uniform particle size distribution while NBDSB presents irregular bulk particles with varying sizes on the surface.

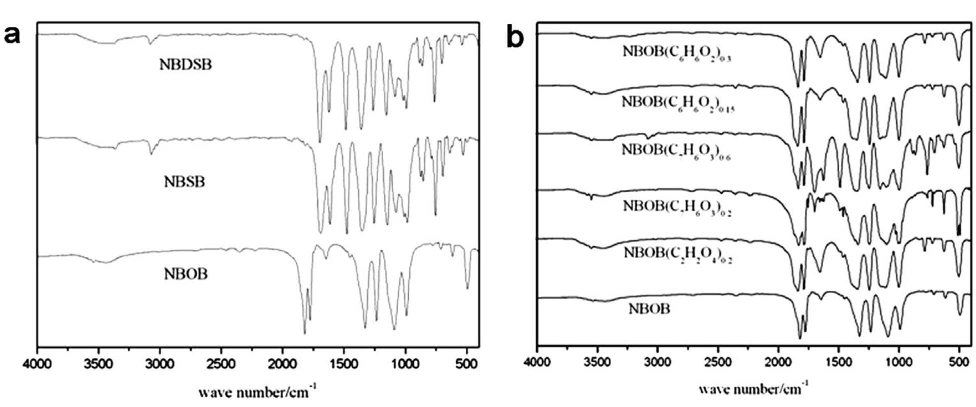


Fig. 1. (a) Comparison of FT-IR spectrum of NBOB, NBSB and NBDSB, (b) FT-IR spectrum of NBOB and NBOB derivates.

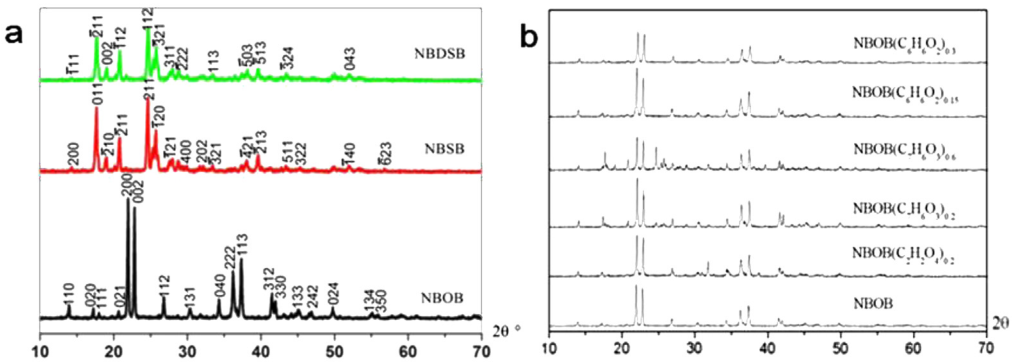


Fig. 2. (a) Comparison of the XRD patterns of NBOB, NBSB and NBDSB, (b) NBOB and NBOB derivates.

Fig. 4a shows the TGA curves of NBOB, NBSB and NBDSB in nitrogen atmosphere. It can be seen that the salt decomposition starts to occur at 345, 353 and 304 °C for NBOB, NBSB and NBDSB, respectively. Decomposition temperature of the ligands include oxalate hydrate, salicylic acid and pyrocatechol are about at 175 °C, 135 °C and 110 °C, respectively, so the thermal stability of five sodium salts is higher than the ligands. Accordingly, they display better thermal stability than lithium salts containing these three ligands (such as LBOB, LBBB, LBSB and LBDOB is at 302, 250, 350 and 256 °C, respectively). Among the sodium salts, NBSB exhibits the highest thermal stability, which depends on the conjugate energies of the chelate-type anion with boron.

Fig. 4b shows the TGA curves of NBOB and NBOB$(C_2H_2O_4)_{0.2}$ doped with oxalic acid in nitrogen atmosphere. It can be seen that the salts decomposition starts to occur at 345 and 304 °C for NBOB and NBOB$(C_2H_2O_4)_{0.2}$. The lower decomposition temperature of NBOB$(C_2H_2O_4)_{0.2}$ maybe due to the increase of oxalic acid.

Fig. 4c shows the TGA curves of NBOB, NBOB$(C_7H_6O_3)_{0.2}$ and NBOB$(C_7H_6O_3)_{0.6}$ doped with salicylic acid in nitrogen atmosphere. It can be observed that the salts decomposition starts to occur at 345, 340 and 345 °C for NBOB, NBOB$(C_7H_6O_3)_{0.2}$ and NBOB$(C_7H_6O_3)_{0.6}$. Note that their decomposition temperature is almost the same with a slight difference, indicative of the good thermal stability of salicylic acid salts as well as NBOB.

Fig. 4d shows the TGA curves of NBOB, NBOB$(C_6H_6O_2)_{0.15}$ and NBOB$(C_6H_6O_2)_{0.3}$ doped with pyrocatechol in N_2 atmosphere. It can be revealed that the decomposition temperature of NBOB$(C_6H_6O_2)_{0.15}$ (300 °C) and NBOB$(C_6H_6O_2)_{0.3}$ (320 °C) are much lower than that of NBOB (345 °C). The decrease of thermal stability of NBOB is attributed to the addition of pyrocatechol. Above all, NBOB, NBSB, NBDSB and five NBOB derivates have no obvious weightlessness at less than 300 °C. That is because these samples are synthesized by solid-state sintering reaction, resulting in good thermal stability.

The solubility of NBOB, NBSB, NBDSB and NBOB derivates were further measured. It was found all of the samples are stable in organic electrolyte solutions but they may decompose by hydrolysis and are not soluble in PC, EC or DMC. What is more, they are

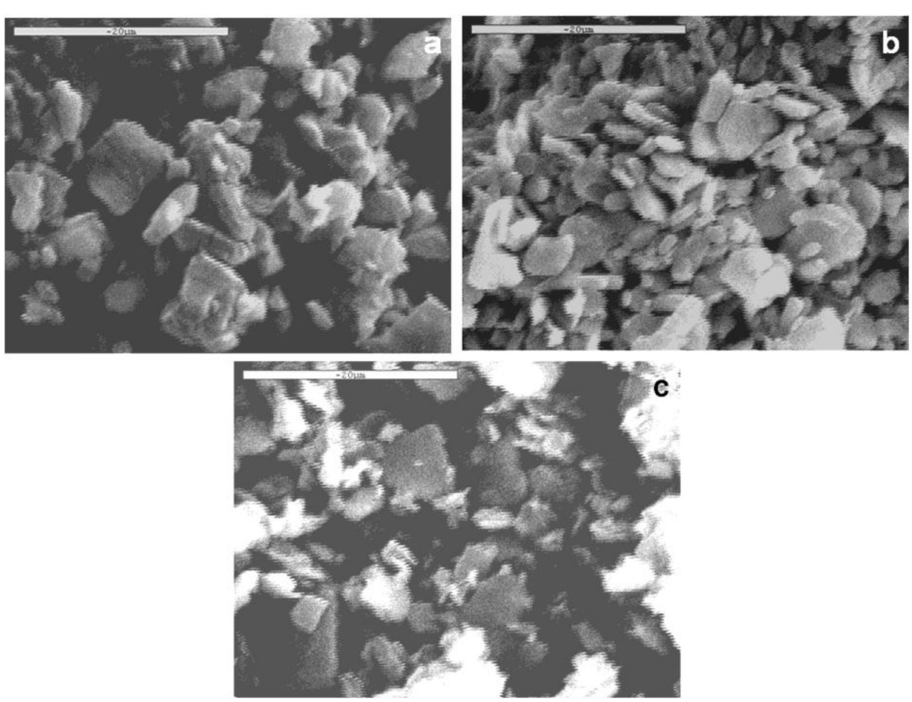


Fig. 3. SEM images of (a) NBOB, (b) NBSB and (c) NBDSB.

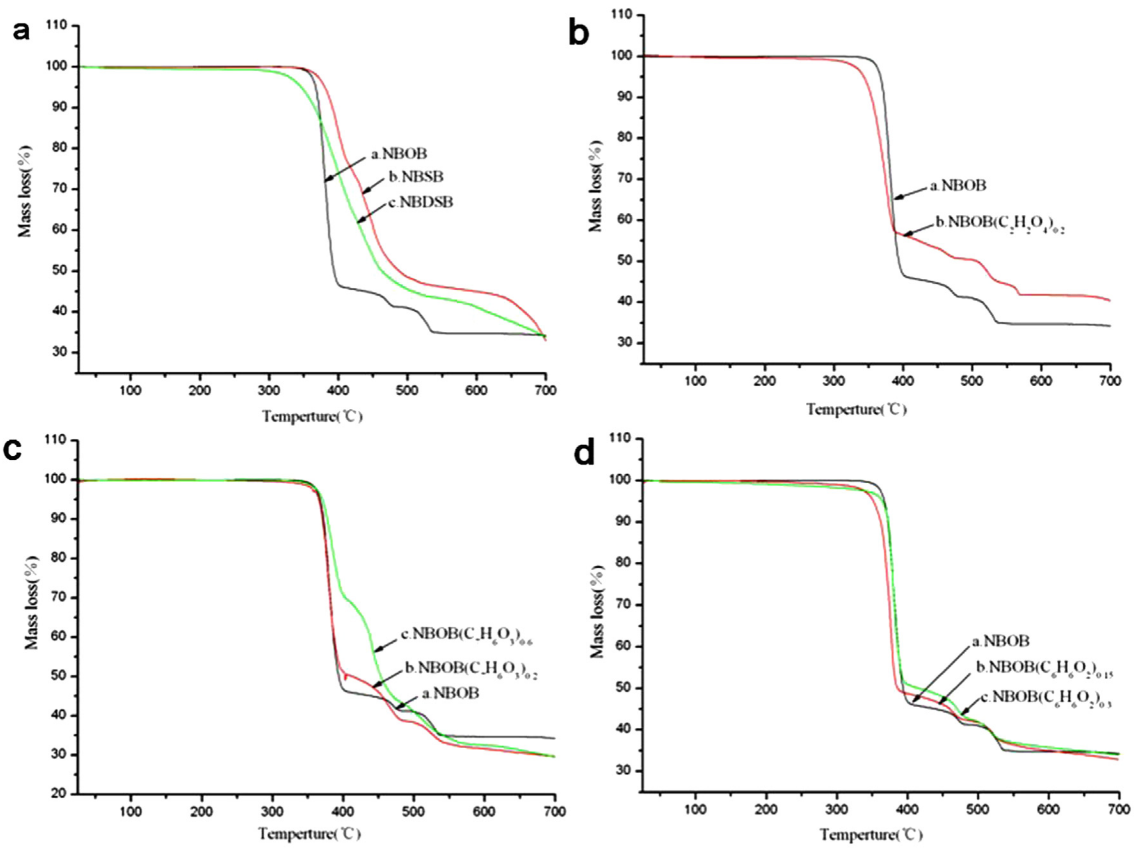


Fig. 4. (a) Comparison of thermal stability from the TGA curves of NBOB, NBSB and NBDSB, (b) NBOB and oxalic acid doped NBOB, (c) NBOB and salicylic acid doped NBOB, (d) NBOB and pyrocatechol doped NBOB at a heating rate of 15 °C min^{-1} in N_2 gas.

moderately soluble in AN, THF or EMC and soluble in DMF or DMSO at room temperature.

Table 1 shows the ionic conductivity of three sodium borate salts in PC, AN, DMF, PC + AN and PC + DMF (volume ratio 1:1) solvents at room temperature. The conductivity in AN solution of 0.025 mol dm^{-3} NBDSB is 1.448×10^{-3} S cm^{-1}. The conductivity in AN solution of 0.025 mol dm^{-3} NBSB is 1.548×10^{-3} S cm^{-1}, indicating that the ions in NBSB solution is highly mobile. The conductivity of 0.025 mol dm^{-3} NBOB in DMF shows conductivity of 1.111×10^{-3} S cm^{-1} at 25 °C. As shown in Table 1, the conductivity of the 0.025 mol dm^{-3} NBSB and NBDSB electrolyte solutions in AN which have higher molecular weight and larger anions is greater than those in the other solvents. Apparently, NBSB and NBDSB are highly dissociating in AN solvent.

Table 1
Specific conductivities (σ, mS cm^{-1}) in different solvents containing 0.025 mol dm^{-3} sodium organoborates at 25 °C.

	PC	AN	DMF	PC + AN	PC + DMF
NBSB	0.256	1.548	0.974	0.599	0.501
NBDSB	0.239	1.448	0.83	0.664	0.414
NBOB	0.071	0.258	1.111	0.160	0.60

Table 2
Specific conductivities(σ, mS cm^{-1}) containing 0.025 mol dm^{-3} NBOB and its derivates in AN and DMF solvents at 25 °C.

	NBOB	$NBOB(C_2H_2O_4)_{0.2}$	$NBOB(C_7H_6O_3)_{0.2}$
AN	0.258	0.366	0.457
DMF	1.111	1.1553	1.268
	$NBOB(C_7H_6O_3)_{0.6}$	$NBOB(C_6H_6O_2)_{0.15}$	$NBOB(C_6H_6O_2)_{0.3}$
AN	0.984	0.237	0.034
DMF	1.156	1.206	0.032

Table 2 shows the conductivity of NBOB and the five NBOB derivates with five organic acids and pyrocatechol in AN and DMF solvents at 25 °C. The result shows that the conductivity of NBOB has been improved obviously after the proper doping of organic acids in AN and DMF solvents, the change is not obvious in DMF, but the change is distinctive in AN. The conductivity of NBOB has been multiplied, especially when salicylic acid doped NBOB. However, the conductivity of NBOB has decreased after small amounts of pyrocatechol doped NBOB in AN solvent. The change is not obvious, but the change is distinctive in AN and DMF solvents when NBOB doped with increased pyrocatechol.

The result confirms that proper amount of organic acids doped NBOB is responsible for improving the solubility of NBOB in common organic solvents. In contrast, the solubility of NBOB becomes lower for pyrocatechol doped NBOB.

4. Conclusions

A new facial strategy was developed to fabricate a series of novel organic-ligand-doped sodium bis(oxalate)-borate complexes of NBOB, NBSB, NBDSB and the five NBOB derivates of NBOB($C_2H_2O_4$)$_{0.2}$, $NBOB(C_7H_6O_3)_{0.2}$, $NBOB(C_7H_6O_3)_{0.6}$, $NBOB(C_6H_6O_2)_{0.15}$ and $NBOB(C_6H_6O_2)_{0.3}$ by solid-state reaction. The resulting salts can be readily soluble in different organic solvents, and have good thermal and chemical stability, and improved ion conductivity. Moreover, these sodium borate salts can possible serve as novel promising electrolytes for sodium ion battery at room temperature, and will open up new opportunities for the use in energy storage devices.

Acknowledgments

We are grateful to the National Natural Science Foundation of China (no. 21171081/B0103), Natural Science Foundation of Liaoning Province, China (no. 201202093), and Foundation of 211 Project for Innovative Talents Training, Liaoning University.

References

[1] J.M. Tarascon, M. Armand, Nature 414 (6861) (2001) 359.
[2] X.P. Ai, H.X. Yang, J. Electrochem. 17 (2) (2011) 123.
[3] J.M. Tarascon, Nat. Chem. 2 (6) (2010) 510.
[4] F. Risacher, B. Fritz, Aquat. Geochem. 15 (2009) 123.
[5] A. Yaksic, J.E. Tilton, Resour. Policy 34 (2009) 185.
[6] Y. Fang, Y. Yao, J.T. Kummer, J. Inorg. Nucl. Chem. 29 (9) (1967) 2467.
[7] B.L. Ellis, L.F. Nazar, Curr. Opin. Solid State Mater. Sci. 16 (4) (2012) 168.
[8] D.A. Stevens, J.R. Dahn, J. Electrochem. Soc. 147 (4) (2000) 1271.
[9] D.A. Stevens, J.R. Dahn, J. Electrochem. Soc. 147 (12) (2000) 4428.
[10] R. Alcantara, J.M. J-Metaeos, P. Lavela, J.L. Tirado, Electrochem. Commun. 3 (4) (2001) 639.
[11] S. Tepavcevic, H. Xiong, R. Vojislav, V.R. Stamenkovic, X.B. Zuo, M. Balasubramanian, V.B. Prakapenka, C.S. Johnson, T. Rajh, ACS Nano 6 (1) (2012) 530.
[12] R. Alcantara, J.M. Jimenez Mateos, J.L. Tirado, J. Electrochem. Soc. 149 (2) (2002) A201.
[13] B.K. Guo, Central South Industry University Press, Changsha, 2000, p. 314.
[14] S.G. Meibuhr, J. Electrochem. Soc. 117 (1) (1970) 56.
[15] L.A. Dominey, V.R. Koch, T.J. Blakley, Electrochim. Acta 37 (9) (1992) 1551.
[16] M. Ue, J. Electrochem. Soc. 141 (12) (1994) 3336.
[17] L.J. Krause, W. Lamanna, J. Summerfield, M. Engle, G. Korba, R. Loch, R. Atanasoski, J. Power Sources 68 (2) (1997) 320.
[18] K.M. Abraham, J.L. Goldman, D.L. Natwig, J. Electrochem. Soc. 129 (11) (1982) 2404.
[19] J. Barthel, R. Buestrich, H.J. Gores, M. Schmidt, M. Wuhr, J. Electrochem. Soc. 144 (11) (1997) 3866.
[20] M. Handa, M. Suzuki, J. Suzuki, H. Kanematsu, Y. Sasaki, Electrochem. Solid-State Lett. 2 (2) (1999) 60.
[21] A. Webber, J. Electrochem. Soc. 138 (9) (1991) 2586.
[22] J.S. Gnanaraj, E. Zinigrad, M.D. Levi, D. Aurbach, M. Schmidt, J. Power Sources 119 (2003) 799.
[23] B.T. Yu, W.H. Qiu, F.S. Li, L.F. Li, J. Power Sources 174 (2) (2007) 1012.
[24] B.T. Yu, W.H. Qiu, F.S. Li, G.X. Xu, Electrochem. Solid State Lett. 9 (1) (2006) A1.
[25] L.Z. Fan, T.F. Xing, R. Awan, W.H. Qiu, Ionics 17 (6) (2011) 491.
[26] T. Yu, H.M. Zhang, X.L. Xu, X.L. Cui, L.P. Mao, Adv. Mater. Mater. Process. 1–3 (652–654) (2013) 896.
[27] K. Xu, S.S. Zhang, B.A. Poese, T.R. Jow, Electrochem. Solid State Lett. 5 (11) (2002) A259.
[28] J. Reiter, R. Dominko, M. Nadherna, I. Jakubec, J. Power Sources 189 (1) (2009) 133.
[29] Z.M. Xue, J.F. Zhao, J. Ding, C.H. Chen, J. Power Sources 195 (3) (2010) 853.
[30] P.Y. Zavalij, S. Yang, M.S. Whittingham, Acta Crystallogr. Sect. B Sci. 59 (2003) 753.
[31] Z.M. Xue, K.N. Wu, B. Liu, C.H. Chen, J. Power Sources 171 (2007) 944.
[32] X.Y. Li, Z.M. Xue, J.F. Zhao, C.H. Chen, J. Power Sources 235 (2013) 274.
[33] J. Barthel, M. Wuhr, R. Buestrich, H.J. Gores, J. Electrochem. Soc. 142 (8) (1995) 2527.
[34] J.W. Visser, J. Appl. Crystallogr. 2 (1969) 89.